

(1) 黎曼积分. (可积性)

定义: 黎曼和 $\Rightarrow$ 达布上和, 下和

$\Rightarrow \begin{cases} \text{上积分} \\ \text{下积分} \end{cases} \Rightarrow$ 可积充分必要条件.

应用定积分定义求一些极限

$$\textcircled{1} \lim_{n \rightarrow \infty} \frac{1}{n} (\sin \frac{\pi}{n} + \sin \frac{2}{n} \pi + \dots + \sin \frac{n}{n} \pi).$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n \sin \xi_i \pi \Delta x = \int_0^1 \sin \pi x dx.$$

$$\textcircled{2} \lim_{n \rightarrow \infty} n^{-\frac{3}{2}} (\sqrt{n+1} + \sqrt{n+2} + \dots + \sqrt{n+n})$$

(2) 不定积分. $\begin{cases} \text{换元法 (有理化)} \\ \text{分部积分法. (注意导数比函数简单的)} \end{cases}$

$$\sqrt{ax^2+bx+c} = \sqrt{(\quad)^2 - \quad}$$

两类函数. $\begin{cases} \text{分式有理} \begin{cases} \text{真分式} \\ \text{假分式} \end{cases} \\ \text{三角有理} \end{cases}$

$$\textcircled{1} \text{分式有理: } \frac{P_n(x)}{Q_n(x)} = \frac{1}{(x+a)} + \frac{1}{(x+a)^2} + \frac{Ax+B}{x^2+px+q} + \frac{Ax+B}{(x^2+1)^2}$$

$$\text{P13. } \frac{1}{(x+1)(x+2)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

$$\frac{x^3+1}{x^3+5x^2+6x} = \frac{x^3+5x^2+6x - 5x^2-6x+1}{x^3+5x^2+6x} = 1 - \frac{5x^2+6x-1}{x(x^2+2)(x+3)}$$

② 三角有理

- 万能公式: $t = \tan \frac{x}{2}$

- $t = \sin x, \cos x, \tan x.$

$$\int \frac{\sin^4 x}{\cos^3 x} dx = \int \frac{\sin^4 x}{\cos^4 x} d \sin x = \int \frac{\sin^4 x}{(1-\sin^2 x)^2} d \sin x = \int \frac{t^4}{(1-t^2)^2} dt.$$

$$\int \frac{\sin^4 x}{\cos^4 x} dx = \int \tan^4 x dx = \int \tan^2 x (\sec^2 x - 1) = \int \tan^2 x d \tan x - \int (\sec^2 x - 1) dx.$$

都是三角有理.

$$\int \frac{\sin x}{\sin^2 x \cos^2 x} dx = - \int \frac{1}{(1-\cos^2 x) \cos^2 x} d\cos x = - \int \frac{1}{(1-t^2)t^2} dt.$$

$$\int \frac{\sin^2 x}{1+\sin^2 x} dx = \int \frac{\sin^2 x}{\csc^2 x + 1} \csc^2 x dx = - \int \frac{\sin^2 x}{\csc^2 x + 1} d\cot x$$

$$= - \int \frac{1}{\cot^2 x + 2} d\cot x.$$

$$\int \frac{1}{5+4\sin x} dx \quad (\text{万能公式})$$

$$\int \frac{\sqrt{x+1} - \sqrt{x-1}}{\sqrt{x+1} + \sqrt{x-1}} dx, \quad \text{令 } \sqrt{\frac{x+1}{x-1}} = t. \Rightarrow t^2$$

$$= \int \frac{t-1}{t+1} () dt.$$

$$\int (\arccos x)^2 dx. \quad \text{令 } \arccos x = t. \quad x = \cos t.$$

$$= - \int t^2 \sin t dt. \quad (\text{分部积分})$$

$$\int x \tan^2 x dx = \int x (\sec^2 x - 1) dx.$$

$$\int \frac{x}{\sin^2 x} dx = - \int x d\cot x$$

不定积分.

1. 尽量产生和式.

① 积化和差. eg: $\int \sin 5x \cos 3x dx.$

② 降幂公式:

③ 裂项.

④ 尽量化简分母, 分子拆开成多个和式:

$$\int \frac{1}{1-\sin x} dx = \int \frac{1+\sin x}{\cos^2 x} dx = \int \sec x dx + \int \frac{1}{\cos^2 x} d\cos x.$$

⑤ 三角恒等式; 辅助角. $\sin + \cos = \sqrt{2} \sin(x + \frac{\pi}{4})$

$$\sin^2 x + \cos^2 x = 1. \quad \sec^2 x - \tan^2 x = 1. \quad \csc^2 x - \cot^2 x = 1.$$

(不仅用于拆和式, 也用于换名).

2. 记住几个常用积分.

$$d(\sec x) = \tan x \sec x dx. \Rightarrow \int \tan x \sec x = \sec x + C$$

$$d(\csc x) = -\cot x \csc x dx \Rightarrow \int \cot x \csc x = -\csc x + C$$

$$d(\cot x) = -\csc^2 x \Rightarrow \int \csc^2 x = -\cot x + C.$$

$$\int \tan x = \ln |\cos x| + C. \quad \int \cot x = \ln |\sin x| + C.$$

$$\int \sec x = \ln |\sec x + \tan x| + C. \quad \int \csc x = \ln |\csc x - \cot x| + C.$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \frac{x}{a} + C.$$

$$\int \frac{1}{\sqrt{x^2 \pm a}} dx = \ln |x + \sqrt{x^2 \pm a}| + C.$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C.$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C.$$

$$\int \sqrt{a^2 - x^2} dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{a} + C.$$

$$\int \sqrt{x^2 \pm a^2} dx = \frac{1}{2} (x \sqrt{x^2 \pm a^2} \pm a^2 \ln |x + \sqrt{x^2 \pm a^2}|) + C.$$

3. $\int \sin ax \sin bx dx$ —— 积化和差.

4. 变换分子.

eg 1. $\int \frac{1}{\sin(x+a) \sin(x+b)} dx$. 其中 $\sin(a-b) \neq 0$.

$$= \frac{1}{\sin(a-b)} \int \frac{\sin(a-b) \overset{x+a-(x+b)}{\sin(x+a) \sin(x+b)}}{\sin(x+a) \sin(x+b)} = \frac{1}{\sin(a-b)} \int \frac{\cos(x+b)}{\sin(x+b)} - \frac{\cos(x+a)}{\sin(x+a)} dx$$

$$= \frac{1}{\sin(a-b)} [\cancel{\ln |\sin(x+b)|} - \cancel{\ln |\sin(x+a)|}] + C.$$

eg 2. $\int \frac{1}{\sin(x+3) \cos(x+5)} dx$.

$$= \frac{1}{\cos 2} \int \frac{\cos(x+5) - (x+3)}{\sin(x+3) \cos(x+5)} dx = \frac{1}{\cos 2} \int \left(\frac{\cos(x+3)}{\sin(x+3)} + \frac{\cos(x+5)}{\sin(x+5)} \right) dx.$$

5. 组合法:

I 不好求, J 不好求, 但 $I+J$, $I-J$ 都好求.

eg: $\int \frac{\cos^2 x}{\sin x + \cos x} dx = I.$

$$J = \int \frac{\sin^2 x}{\sin x + \cos x} dx$$

$$\begin{aligned} I+J &= \int \frac{1}{\sin x + \cos x} dx = \frac{\sqrt{2}}{2} \int \frac{1}{\sin(x+\frac{\pi}{4})} d(x+\frac{\pi}{4}) \\ &= \frac{\sqrt{2}}{2} \ln \left| \csc(x+\frac{\pi}{4}) - \cot(x+\frac{\pi}{4}) \right| + C. \end{aligned}$$

$$I-J = \int (\cos x - \sin x) dx = \sin x + \cos x + C.$$

$$I = \frac{1}{2}(\sin x + \cos x) + \frac{\sqrt{2}}{4} \ln \left| \csc(x+\frac{\pi}{4}) - \cot(x+\frac{\pi}{4}) \right| + C.$$

(3) 定积分.

① 区间再现 (最后一招)

往往简单.

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx = \frac{1}{2} \int_a^b (f(x) + f(a+b-x)) dx.$$

eg 1. 求 $\int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^2} dx$.

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^2} + \frac{1}{1+(\cot x)^2} = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{4}$$

$\Rightarrow$ 变 $\int_0^{+\infty} \frac{1}{(1+x^2)(1+x^2)} dx$.

$$= \int_0^{+\infty} \frac{1}{1+x^2} d \arctan x = \int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan t)^2} dt.$$

eg 2. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\arctan e^x) \sin^2 x dx$.

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x (\arctan e^x + \arctan e^{-x}) dx.$$

积累: $\arctan x + \arctan \frac{1}{x} = \begin{cases} \frac{\pi}{2}, & x > 0 \\ -\frac{\pi}{2}, & x < 0. \end{cases}$

② 华里士 (点火) 公式.

$$\int_0^{\frac{\pi}{2}} \sin^n x dx \quad \left\{ \begin{array}{l} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2}, \quad n \text{ 是偶数} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{2}{3} \text{ 发射失败. } \quad n \text{ 为奇数.} \end{array} \right.$$

11 区间再现

$$\int_0^{\frac{\pi}{2}} \cos^n x dx$$

③ 性质.

线性性. 保序性. 对应域可加, 中值定理. 奇偶函数. 周期函数.

④ 定积分计算 / 证明

换元. 分部积分. 产生递推公式.

应用:

意它们的使用范围和条件,尤其是采用参数方程时积分上下限的取法.

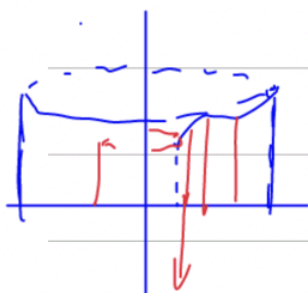
	直角坐标显式方程 $y=f(x), x \in [a, b]$	直角坐标参数方程 $\begin{cases} x=x(t), \\ y=y(t), \end{cases} t \in [T_1, T_2]$	极坐标方程 $r=r(\theta), \theta \in [\alpha, \beta]$
平面图 形面积	$\int_a^b f(x) dx$	$\int_{T_1}^{T_2} y(t)x'(t) dt$	$\frac{1}{2} \int_{\alpha}^{\beta} r^2(\theta) d\theta$
弧长的 微分	$dl = \sqrt{1+[f'(x)]^2} dx$	$dl = \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$	$dl = \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$
曲线 弧长	$\int_a^b \sqrt{1+[f'(x)]^2} dx$	$\int_{T_1}^{T_2} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$	$\int_{\alpha}^{\beta} \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$
旋转体 体积	$\pi \int_a^b [f(x)]^2 dx$	$\pi \int_{T_1}^{T_2} y^2(t) x'(t) dt$	$\frac{2}{3} \pi \int_{\alpha}^{\beta} r^3(\theta) \sin \theta d\theta$
旋转曲 面面积	$2\pi \int_a^b f(x) \sqrt{1+[f'(x)]^2} dx$	$2\pi \int_{T_1}^{T_2} y(t) \sqrt{x'^2(t) + y'^2(t)} dt$	$2\pi \int_{\alpha}^{\beta} r(\theta) \sin \theta \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$

曲线的曲率

$$\text{曲率: } K = \left| \frac{d\varphi}{ds} \right| = \frac{|x'(t)y''(t) - x''(t)y'(t)|}{(x'(t)^2 + y'(t)^2)^{\frac{3}{2}}}$$

$$= \frac{|y''(x)|}{(1 + (y'(x))^2)^{\frac{3}{2}}}$$

→ y 旋转体体积



剥开一层

$$V_y = \int_a^b 2\pi x y(x) dx$$

(4) 变限积分

$$\left(\int_{\alpha(x)}^{\beta(x)} f(t) dt\right)' = f(\beta(x))\beta'(x) - f(\alpha(x))\alpha'(x)$$

条件: ①. $\alpha(x)$ $\beta(x)$ 可导.

②. $f(x)$ 连续.

5. 广义积分. $\left\{ \begin{array}{l} \text{无界区间} \\ \updownarrow \\ \text{无界函数} \end{array} \right\} \left\{ \begin{array}{l} \text{敛散判断} \\ \text{计算} \end{array} \right.$

(1) 概念. 定义. Cauchy 准则.

收敛性判断 $\left\{ \begin{array}{l} \text{非负函数} \left\{ \begin{array}{l} \text{比较判别} \\ \text{极限形式} \\ \text{比阶判别: } \left\{ \frac{1}{x^p} \right\}, \left\{ \frac{1}{x^p \ln x} \right\} \end{array} \right. \\ \text{一般函数} \left\{ \begin{array}{l} \text{条件} \\ \text{绝对} \end{array} \right\} \left\{ \begin{array}{l} \text{狄利克雷判别} \\ \text{阿贝尔判别} \end{array} \right. \end{array} \right.$

(2) 瑕积分. $\int_0^a f(x) dx$ (0 为奇点)

$\left\{ \begin{array}{l} \text{尺度: } \left\{ \frac{1}{x^p} \right\}, \left\{ \frac{1}{x^p \ln x} \right\} \\ \text{狄利克雷, 阿贝尔} \end{array} \right.$

(3) 计算.

复合型, 必须都发散.

* $\int_1^{+\infty} \frac{\sin x}{x^p} dx. \left\{ \begin{array}{l} p \in (0, 1] \text{ 条件收敛, (狄利克雷)} \\ p > 1. \text{ 绝对收敛. (} \end{array} \right.$

$\left| \frac{\sin x}{x^p} \right|, p > 1$ 时. $\left| \frac{\sin x}{x^p} \right| < \frac{1}{x^p}$. 收敛.

$p \leq 1$ 时. $\left| \frac{\sin x}{x^p} \right| \geq \frac{(\sin x)^2}{x^p} = \frac{1}{2x^p} - \frac{\cos 2x}{2x^p}$
发散.

5. 微分方程.

初值问题解存在唯一性定理

① $y' = f(x, y)$. f 连续.

② $\exists L > 0. \forall y_1, y_2, |f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|$

(1) 微分方程

- 阶
 - 变量可分离
 - 可化为齐次. $\frac{dy}{dx} = f(\frac{y}{x})$. 令 $u = \frac{y}{x}$.
 - $y' = (x+y-3)^2$. $\frac{dy}{dx} = \frac{x+y-1}{x-y+2}$ 等.
 - 阶线性方程:
 - $y' + p(x)y = f(x) \longrightarrow y = e^{-\int p(x)dx} (\int f(x)e^{\int p(x)dx} dx + C)$
 - $y' + p(x)y = f(x)y^\alpha \longrightarrow \frac{1}{1-\alpha}(y^{1-\alpha})' + p(x)y^{1-\alpha} = f(x)$
- $\geq$ 阶, $y'' =$
 - 缺 $y, y' = f(x)$. 连积
 - 缺 $x = f(y, y')$ 令 $p = \frac{dy}{dx}$. $\therefore p' = \frac{dp}{dy} \cdot y'$
 - 缺 $y = f(y', x)$ 令 $p = \frac{dy}{dx}$. $p' = y''$
 - 缺 y'

↓
- 高阶线性方程.
 - 解的结构
 - 常系数求解.
 - 欧拉方程, 令 $t = \ln|x|$

(2) 应用

几何

积分方程: eg1: $f(x) = e^x + \int_0^x f(t)dt$. $f \in C$.

可导 $\Leftarrow$ 可导 + 可导

① 能求导吗? 可, $f'(x) = e^x + f(x)$

且有初值 $f(0) = 0$.



※ 变限积分.

eg 1: $f(x) = e^x + \int_0^x f(x-t) dt.$

令 $x-t = u$. $\therefore f(x) = e^x + \int_x^0 f(u) d(u)$

定积分证明:

(1) 估值定理.

$f(x) \in C[a, b]$. $M, m = \sup, \inf.$

则 $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$

(2) 构造变上限积分证明不等式.

变限积分是连接定积分与不等积分的桥梁.

将问题 $\longrightarrow$ 函数问题.

eg 1. 若 $f(x) \in C[a, b]$. 且 $f(x) \uparrow$. 求证:

$$\int_a^b x f(x) dx \geq \frac{a+b}{2} \int_a^b f(x) dx.$$

pf: 即 $2 \int_a^b x f(x) dx - (a+b) \int_a^b f(x) dx.$

令 $F(x) = 2 \int_a^x t f(t) dt - (a+x) \int_a^x f(t) dt.$

于是 $F(a) = 0$. 即证: $F(b) \geq 0$.

$$F'(x) = 2x f(x) - \int_a^x f(t) dt - (a+x) f(x)$$

$$= x f(x) - a f(x) - \int_a^x f(t) dt$$

① 首先. 能否继续求导? $\times$. 因 $f(x)$ 未必可导. $!!$

$$\begin{aligned} \text{于是由积分中值. } F'(x) &= (x-a)f(x) - (x-a)f(\xi) \\ &= (x-a)(f(x) - f(\xi)) > 0. \end{aligned}$$

$\therefore F(x) \uparrow, \therefore F(b) \geq F(a) = 0.$ 毕:

(3) 常用不等式

eg 2. (柯西不等式) $[\int_a^b f(x)g(x)dx]^2 \leq \int_a^b f^2(x)dx \int_a^b g^2(x)dx$

$$\text{令 } F(x) = [\int_a^x f(t)g(t)dx]^2 - \int_a^x f^2(t)dt \int_a^x g^2(t)dt.$$

(这种证明要求 $f(x), g(x)$ 均连续. 但柯西不等式实际只需

$f, g \in R[a, b]$ 即可).

$$\begin{aligned} F'(x) &= 2\int_a^x f(t)g(t)dt \cdot f(x)g(x) - f^2(x)\int_a^x g^2(t)dt - g^2(x)\int_a^x f^2(t)dt \\ &= \int_a^x 2f(x)g(x)f(t)g(t)dt - \int_a^x f^2(x)g^2(t)dt - \int_a^x g^2(x)f^2(t)dt \\ &= -\int_a^x (f(x)g(t) - f(t)g(x))^2 dt < 0. \end{aligned}$$

$\therefore F(x) \downarrow, \therefore F(b) \leq F(a) = 0$

应用: 若 $f \in C[0, 1], f(x) > 0$. 求证 $\frac{\int_0^1 f(x)^3 dx}{\int_0^1 f^2(x) dx} \geq \frac{\int_0^1 f^2(x) dx}{\int_0^1 f(x) dx}.$

$$\begin{aligned} \text{即证 } \int_0^1 f^3(x) dx \int_0^1 f(x) dx &\geq [\int_0^1 f^2(x) dx]^2 \\ &= \int_0^1 (f^{\frac{3}{2}}(x))^2 dx \int_0^1 (f^{\frac{1}{2}}(x))^2 dx \end{aligned}$$

(2) (闵可夫斯基不等式)

$$\int_a^b [f(x) + g(x)]^2 dx \leq [\int_a^b f^2(x) dx]^{\frac{1}{2}} + [\int_a^b g^2(x) dx]^{\frac{1}{2}}]^2$$

证明用柯西

4. 与周期性有关.

eg1: $f(x)$ 是周期为 2 的连续函数, 证明:

$$\textcircled{1} \forall t \in \mathbb{R}. \int_t^{t+2} f(x) dx = \int_0^2 f(x) dx.$$

$$\textcircled{2} g(x) = \int_0^x [2f(t) - \int_t^{t+2} f(s) ds] dt \text{ 是周期为 2.}$$

pf. $\textcircled{1}$ 方法: 拆积分区间:

$$\int_t^{t+2} f(x) dx = \int_t^0 f(x) dx + \int_0^2 f(x) dx + \int_2^{t+2} f(x) dx.$$

$$\text{即证 } \int_t^0 f(x) dx + \int_2^{t+2} f(x) dx = 0. \quad \text{令 } u = x - 2.$$

$$\therefore \int_2^{t+2} f(x) dx = \int_0^t f(u+2) du = \int_0^t f(u) du = -\int_t^0 f(x) dx. \quad \text{证毕.}$$

$$\begin{aligned} \textcircled{2} g(x) &= 2 \int_0^x f(t) dt - \int_0^x \left(\int_t^{t+2} f(s) ds \right) dt \\ &= 2 \int f(t) \Big|_0^x - \int \left(\int_t^{t+2} f(s) ds \right) \Big|_0^x \rightarrow \text{由 } \textcircled{1} = \\ &= 2 \int f(x) - 2 \int f(0) - \int \int_x^{x+2} f(s) ds + \int \int_0^2 f(s) ds. = 0 \\ g(x+2) &= 2 \int f(x+2) - 2 \int f(0) + \\ \therefore g(x) &= g(x+2) \end{aligned}$$

$$\begin{aligned} \textcircled{2} \text{法二: } g(x+2) &= \int_0^{x+2} [2f(t) - \int_t^{t+2} f(s) ds] dt \\ &= \int_0^x [\quad] dt + \int_x^{x+2} [2f(t) - \int_t^{t+2} f(s) ds] dt \end{aligned}$$

$$\begin{aligned} \text{而 } 2 \int_x^{x+2} f(t) dt - \int_x^{x+2} \left(\int_t^{t+2} f(s) ds \right) dt \\ &= 2 \int_0^2 f(t) dt - \int_x^{x+2} \left(\int_0^2 f(s) ds \right) dt \quad \text{无关} \\ &= 2 \int_0^2 f(t) dt - 2 \int_0^2 f(s) ds = 0. \end{aligned}$$

5. 中值定理证明

结合几大中值定理和解微分方程的手段.

注意: $g(x) = \int_a^x f(t) dt$. $g'(x) = f(x)$.

eg. $f(x), g(x)$ 在 $[a, b]$ 连续 且 $\int_a^b f(x) dx = 0$.

求证 $\exists \xi \in (a, b)$, s.t. $f(\xi) + g(\xi) \int_a^\xi f(x) dx = 0$.

证原: 即 $f(x) + g(x) \int_a^x f(t) dt = 0$.

即 $F'(x) + g(x) F(x) = 0$. 一阶线性.

故构造 $G(x) = F(x) e^{\int g(x) dx}$
 $= \int_a^x f(t) dt e^{\int g(x) dx} = \int_a^x f(t) dt e^{\int_a^x g(t) dt}$. 基-原函数

$\therefore G(a) = 0$. $G(b) = 0$

$\therefore \exists \xi$, s.t. $G'(\xi) = 0$.

补-构造: 若: $f(x) + \lambda f'(x) = 0$.

则 $F(x) = f(x) e^{\lambda x}$ ($\lambda =$ 常数; 函数均可)

6. 定积分等式证明.

(1) 利用分部积分.

证明 $\int_a^b f^{(k)}(x) g(x) dx = \int_a^b f(x) g^{(k)}(x) dx$.

利用分部积分. 把 $f^{(k)}(x)$ 一次次放回 d 里, 用公式, 得到

$$\int_a^b f^{(k)}(x) g(x) dx = \underbrace{M(x)}_{=0} + \int_a^b f(x) g^{(k)}(x) dx,$$

eg: $f''(x) \in C[0, 1]$, 且 $f(0) = f(1) = 0$. 证 $\int_0^1 f(x) dx = \frac{1}{2} \int_0^1 x(x-1) f''(x) dx$

p.f. $\int_0^1 f(x) dx = \int_0^1 [\frac{1}{2} x(x-1)]'' f(x) dx$.

$$\begin{aligned} \text{设 } 1 \uparrow &= \int_0^1 f(x) d(x - \frac{1}{2}) = (x - \frac{1}{2}) f(x) \Big|_0^1 - \int_0^1 (x - \frac{1}{2}) f'(x) dx \\ &= \underbrace{(x - \frac{1}{2}) f(x) \Big|_0^1}_0 - \underbrace{\frac{1}{2} x(x-1) f'(x) \Big|_0^1}_0 + \int_0^1 \frac{1}{2} x(x-1) f''(x) dx, \end{aligned}$$

(2) 区间再现.

关于 $\int_a^b f(x)g(x)dx$ 中 $f(x)$ 和 $g(x)$ 的中心对称和周期性.

(3) 含参号问题.

① 零点存在性.

设 $F(x)$ (利用变限积分) $\rightarrow$ 找两点 $\rightarrow F(\xi)=0$.

eg: $f(x) \in C[a, b]$, $f(x) > 0$. 证明 $\exists \xi \in (a, b)$. $\int_a^\xi f(x)dx = \int_\xi^b f(x)dx$.

pf: 令 $F(x) = \int_a^x f(t)dt - \int_x^b f(t)dt$

$$\therefore F(a) = -\int_a^b f(t)dt < 0 \quad F(b) > 0$$

$\therefore \exists \xi \quad F(\xi) = 0$.

② 泰勒展开 + 介值定理.

含有 $\int_a^b f(x)dx$ 和 $f''(\xi)$ (高阶导) 时

某点展开 $\rightarrow$ 积分 $\rightarrow$ 介值定理.

eg: $f(x)$ 在 $[-a, a]$ 二阶可导, $f(0)=0$. 则 $\exists \eta \in [-a, a]$.

使 $a^3 f''(\eta) = 3 \int_{-a}^a f(x)dx$.

在 0 处展开: $f(x) = f'(0)x + \frac{f''(\eta)}{2}x^2$

$$\text{积分: } \int_{-a}^a f(x)dx = \frac{1}{2}f'(0)x^2 \Big|_{-a}^a + \frac{1}{6}f''(\eta)x^3 \Big|_{-a}^a$$

$$= \frac{1}{3}f''(\eta)a^3$$

η 介于 $(0, x)$ 之间.
说明 η 是与 x 相关的.

利用积分中值: 由于 $f''(x) \in C[a, b]$.

故 $f''(x)$ 在 $[a, b]$ 上有最大 M , 小 m .

$$\therefore \exists f''(\eta) \in [m, M]. \quad \int_{-a}^a \frac{1}{2}f''(\eta)x^2 dx = \frac{1}{2}\eta \int_{-a}^a x^2 dx = \frac{1}{3}f''(\eta)a^3$$

证毕.

③ 分部积分 + 积分中值.

含有 $\int_a^b f(x) g(x) dx$. $g(x) \int_a^x f(t) dt$. $f(x) \int_x^b g(t) dt$.

视 $f(x) = \left(\int_a^x f(t) dt \right)'$

则 $\int_a^b f(x) g(x) dx \xrightarrow{\text{分部}} \int_a^b (g'(x) \int_a^x f(t) dt) dx$.

若能用积分中值

↓ 积分中值.

$$\int_a^x f(t) dt \cdot (g(b) - g(a))$$

eg: $f(x), g(x) \in C[a, b]$. $g(x)$ 可导. 且 $g'(x) \geq 0$. 证明: $\exists \xi \in [a, b]$.

使 $\int_a^b f(x) g(x) dx = g(b) \int_a^b f(x) dx + g(a) \int_a^x f(x) dx$.

令 $F(x) = \int_a^x f(t) dt$. $G(x) = \int$

$$\int_a^b f(x) g(x) dx = F(x) g(x) \Big|_a^b - \int_a^b g'(x) F(x) dx$$

$$= g(b) \int_a^b f(x) dx - \int_a^x f(x) dx (g(b) - g(a))$$

$$= g(b) \int_a^b f(x) dx + g(a) \int_a^x f(x) dx.$$

④ 化为微分中值问题.

设 $F(x) = \int_a^x f(t) dt$. 将含积分的等式变得不含积分.

eg: 设 $f, g \in C[a, b]$. 证: $\exists \xi \in (a, b)$. 使 $f(\xi) \int_a^b g(x) dx = g(\xi) \int_a^x f(x) dx$.

pf. 设 $F(x) = \int_a^x f(t) dt$. $G(x) = \int_a^x g(t) dt$.

$$\therefore \Leftrightarrow F'(\xi) (G(b) - G(\xi)) = G'(\xi) (F(\xi) - F(a))$$

$$\text{即 } F'(\xi) G(b) - F'(\xi) G(\xi) = G'(\xi) F(\xi) - G'(\xi) F(a)$$

$$\Rightarrow F'(\xi) G(\xi) + G'(\xi) F(\xi) - F'(\xi) G(b) - G'(\xi) F(a) = 0. \quad (\text{技巧. 把 } \xi \rightarrow x \rightarrow \text{函数}).$$

∴ 需要是某个 $M(x)$ 的导数.

$$\therefore \text{令 } M(x) = F(x) G(x) - F(a) G(x) - G(b) F(x) \quad \therefore M(a) = M(b)$$

∴ $\exists \xi \in (a, b)$, $M'(\xi) = 0$. 证毕.

eg 2. $f \in C[a, b]$, 可导, 证明 $\exists \xi \in (a, b)$, 使

$$f(b) \int_a^b f(x) dx = (b-a) [f'(\xi) \int_a^b f(t) dt + f^2(\xi)]$$

$$F(x) = \int_a^x f(t) dt, \quad F'(b) F(b) = (b-a) [F'(\xi) F(\xi) + F'(\xi)^2]$$

$$\Rightarrow (F'(x) F(x))' = \frac{F'(b) F(b) - F'(a) F(a)}{b-a} \quad \text{拉格朗日}$$

9. 设 $f(x)$ 在 $[1, 2]$ 上连续, 在 $(1, 2)$ 内可导, 且 $f(x) \neq 0$ ($1 < x < 2$),

证明: 存在 $\xi \in (1, 2)$, 使得 $\frac{\ln 2}{\int_1^2 f(t) dt} = \frac{1}{\xi f(\xi)}$

解: ① 设 $\int_1^x f(t) dt = F(x)$

② 变形所证式子 所证式子变为 $\frac{\ln 2}{F(2)} = \frac{1}{\xi f(\xi)}$

③ 运用微分中值定理 分式 + 参数的形式, 考虑柯西中值定理, 看能否构造出 $\frac{F(2)-F(1)}{F(2)-F(1)}$ 形式

由于 $\ln 1 = 0, F(1) = 0 \therefore$ 可以转换为 $\frac{\ln 2 - \ln 1}{F(2) - F(1)} = \frac{1}{\xi f(\xi)}$, 证毕 (运用柯西中值定理证)。

※ 辅助多项式法.

给一堆条件, 求证 $f^{(k)}(x_0) = a$.

方法: 构造 $p(x) = p_k(x)$ 多项式函数, 代已知条件解得定系数
解不全的系数根据需要人为赋值.

最终 罗尔 解决.